Clinical Risk Assessment Report: Patient 98

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 59.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.67x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +7.6%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.06 [Top 10%] (+7.6%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.32 [Bottom 19%] (+5.8%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.79 [Top 25%] (+5.7%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.81 [Top 22%] (+5.5%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 3.74 [Top <1%] (+4.0%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.97 [Top 15%] (+3.3%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.26 [Bottom 4%] (+2.7%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.63 [Top 4%] (+2.5%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)